Sailing the Noise Ocean: A Hardware Agnostic Noise Ablation Study of Quantum-Classical Neural Networks

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INTRODUCTION

- Quantum machine learning (QML) is an emerging paradigm of machine learning (ML) that demonstrates superior efficiency to classical counterparts in pivotal fields such as quantum chemistry.
- In the current Noisy Intermediate-Scale Quantum (NISQ) era, QML is susceptible to various channels of noise that degrades gradient signals.
- We conduct a noise ablation study to identify and quantify the extent of these effects caused by the two major channels of noise:
  - Depolarizing Error
  - Thermal Relaxation Error

BACKGROUND

- QNNs utilize a hybrid quantum-classical ansatz, however the hardware noise present prevents efficient convergence.
- Our study considers two primary noise channels:
  - **Depolarizing error:** Leaves qubits intact with probability *(1-p)* or replaces it with the maximally mixed state with probability *p*.
  - **Thermal Relaxation error:** The physical loss of energy of a quantum state over time.

DATA

- We use a stratified random sampling of 100 images from MNIST [2], to classify the digit 0 vs 1.
- We use 300 validation, and 500 test images.

METHOD

- We take the median noise present in each channel of the Rensselaer machine's noise model, and parametrize it with 2 parameters α and γ, with α corresponding to Depolarizing Error and γ corresponding to Thermal Relaxation error.
- Below is the specific architecture we evaluate:

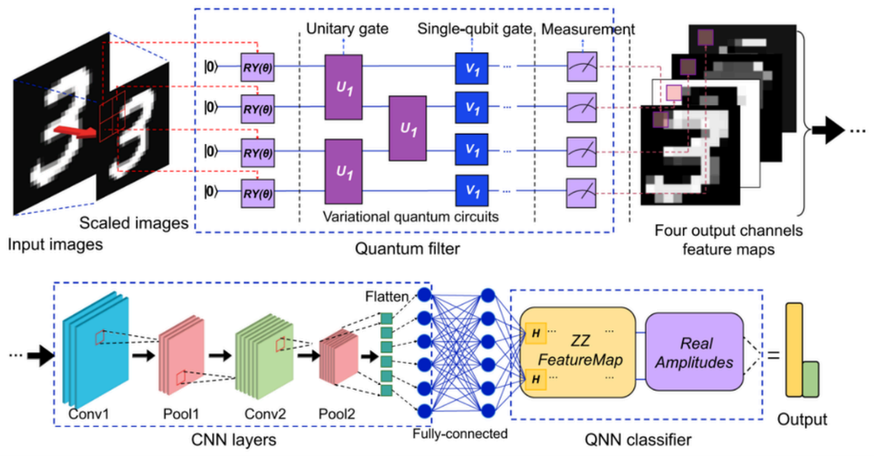


Fig. 1. We measure the tolerance of a Hybrid Quantum Classical Convolutional Network Proposed in [1].

RESULTS

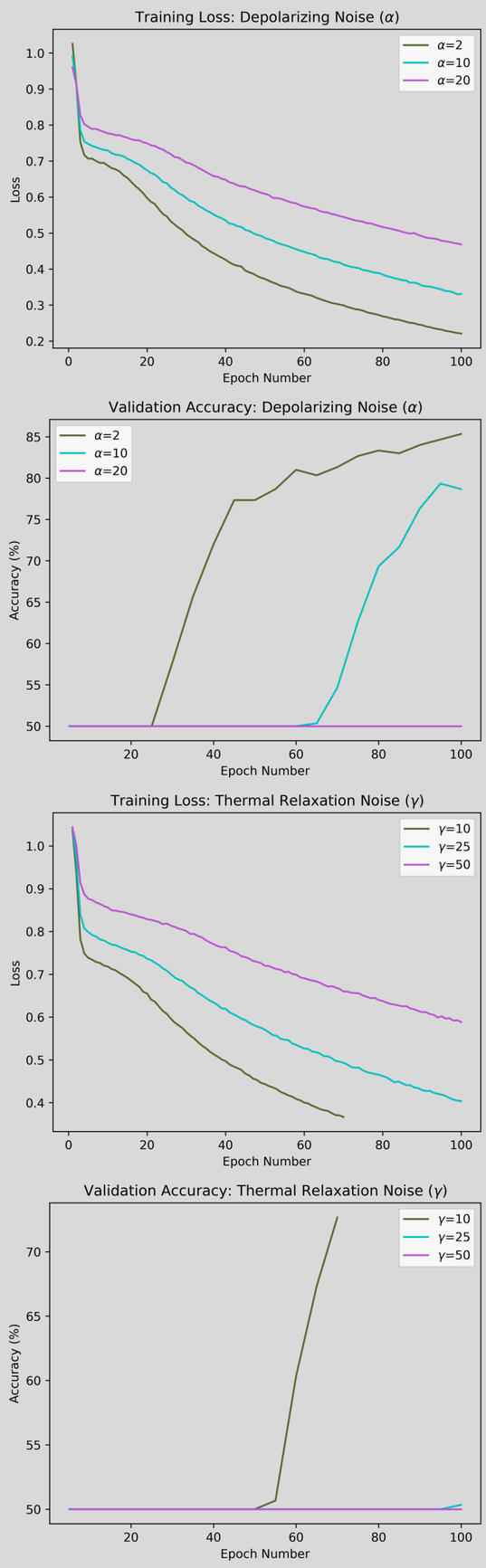


Fig. 2, Fig. 3. Loss and Validation for the depolarizing noise channel.

- Depolarizing noise scrambles the quantum state of each of the qubits themselves.
- At $\alpha = 20$, the signal is lost, and the model is unable to converge.

Fig. 4, Fig 5. Loss and Validation for the thermal relaxation noise channel.

- Thermal Relaxation causes the states of qubits to decay.
- At $\gamma = 25$ the signal is mostly gone, but the slight increase of accuracy at the end signals that given more epochs, the model could converge.
- At $\gamma = 50$, the signal is completely gone.

FURTHER RESULTS

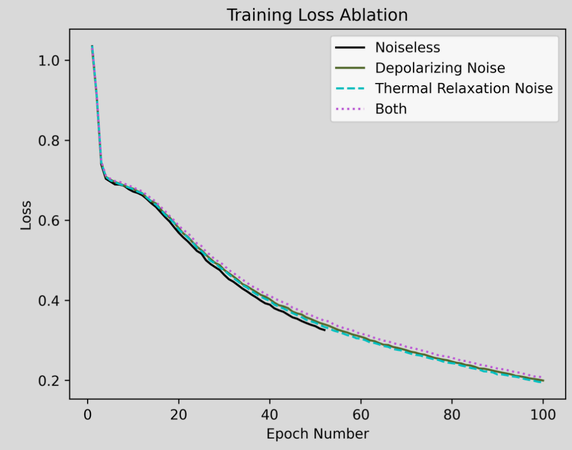


Fig. 6. Training loss with only certain noise channels present. Note that we set the parameter (α or γ) to 1 here.

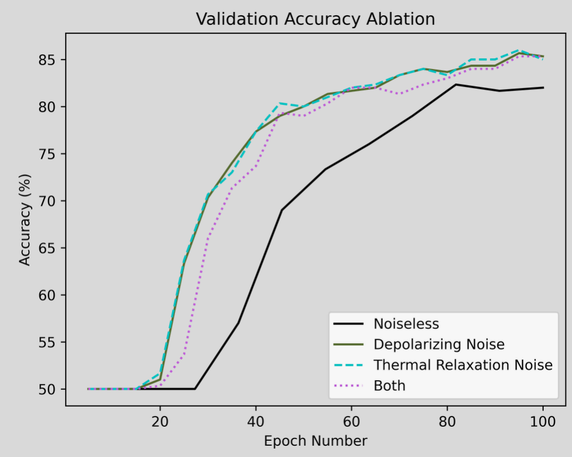


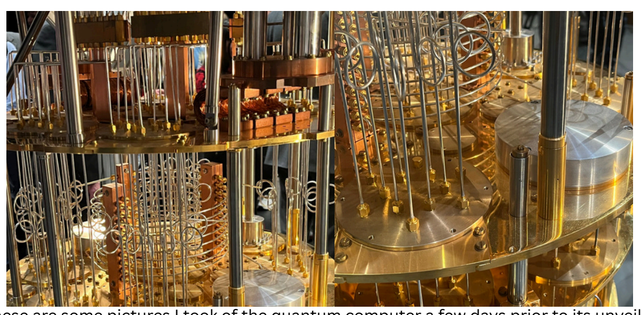
Fig. 7. Validation accuracy with only certain noise channels present.

Noise Profile	Test Accuracy
Depolarizing	**89.4%**
Thermal	**89.0%**
Both	**88.4%**

Table. 1. Test accuracy with only certain noise channels present.

CONCLUSION

- We have analyzed the effects of 2 major channels of hardware noise.
- Too much of these channels prevent the model from converging.
- The QCQ-CNN proposed in [1] can support the physical noise present in the current hardware, as evidenced by the ablation study.
- Subsequent research can explore the effects of these noise channels on different models, and can transition to training on real quantum hardware.



These are some pictures I took of the quantum computer a few days prior to its unveiling.

REFERENCES

- [1] C. Long, M. Huang, X. Ye, Y. Futamura, and T. Sakurai, "Hybrid quantum-classical-quantum convolutional neural networks," Scientific Reports, vol. 15, no. 1, Aug. 2025, doi: https://doi.org/10.1038/s41598-025-13417-1.
- [2] Y. Lecun, L Eon Bottou, Y. Bengio, and Patrick Haaner Abstract|, "Gradient-Based Learning Applied to Document Recognition," PROC. OF THE IEEE, 1998, Available: http://vision.stanford.edu/cs598_spring07/papers/Lecun98.pdf